AGILE DEVELOPMENT AND DEVOPS

LAB ASSESSMENT-2

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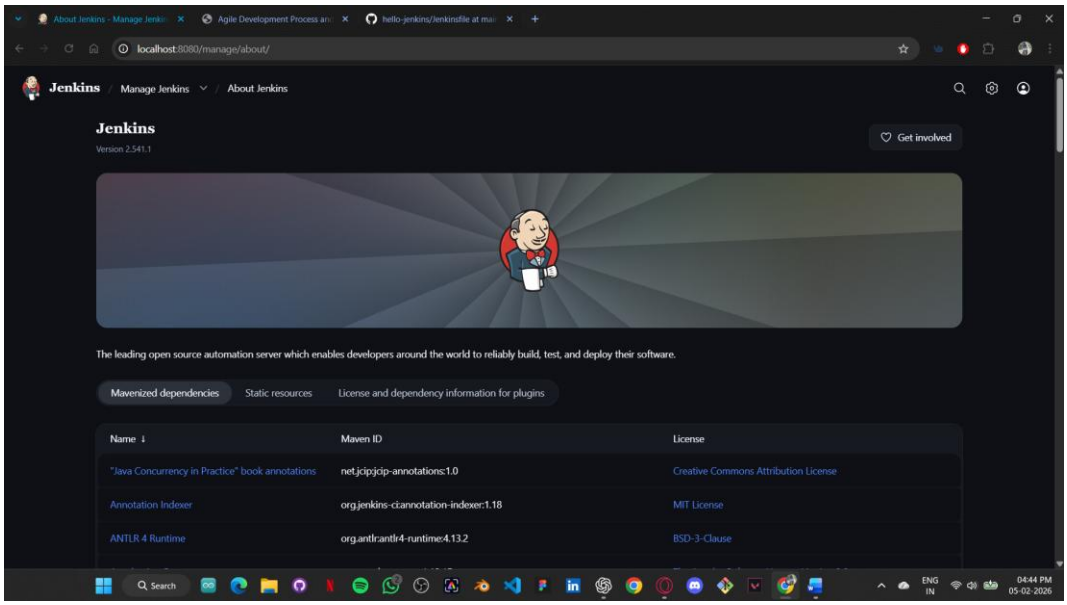
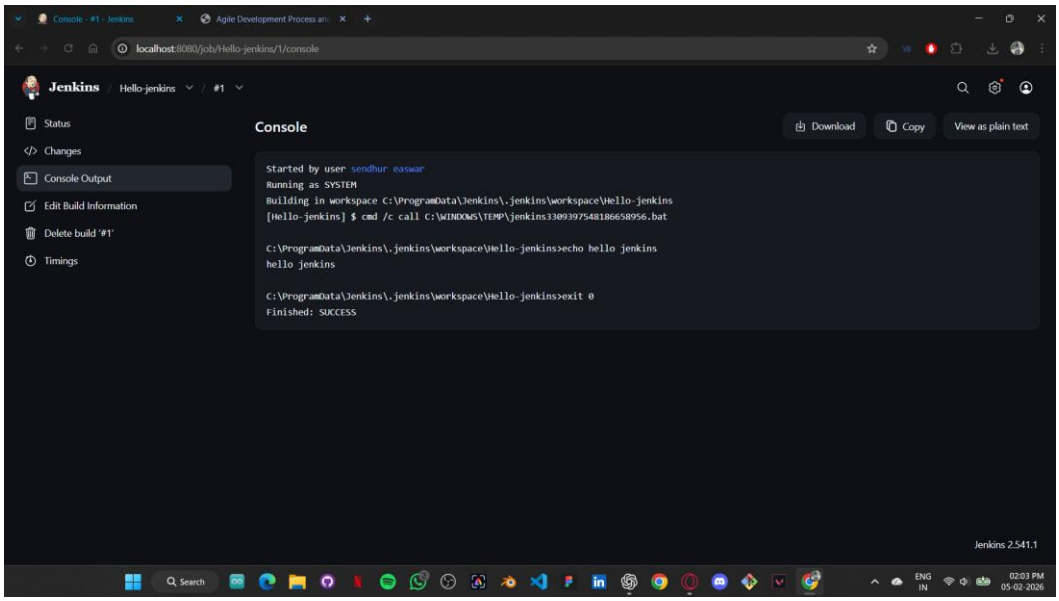
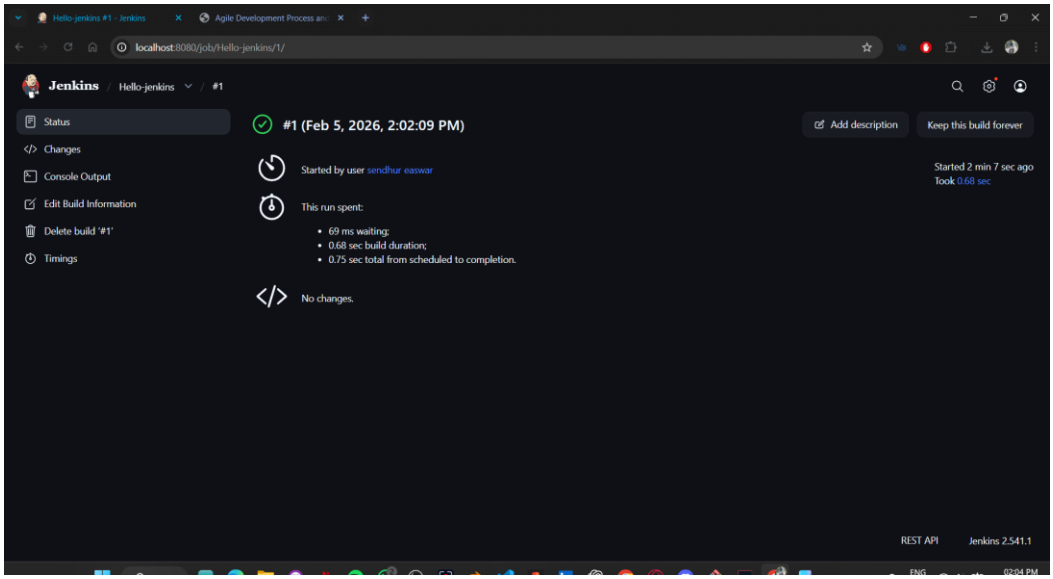
REGNO:23MIS0306

TASK 1: Jenkins Familiarization Objective: Understand Jenkins UI and basic navigation Tasks: 1. Open Jenkins Dashboard in browser 2. Identify: o Dashboard o Manage Jenkins o New Item o Build History 3. Check Jenkins version Expected Output: • Screenshot or note of Jenkins version

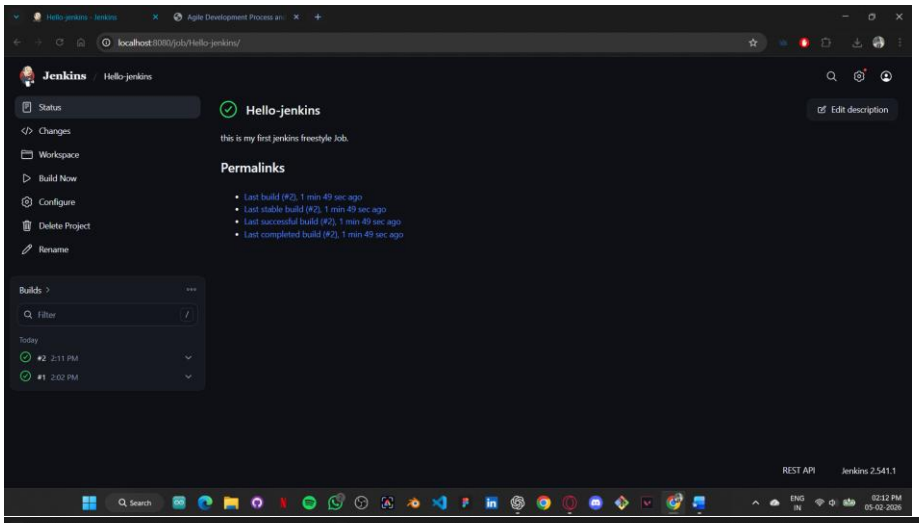


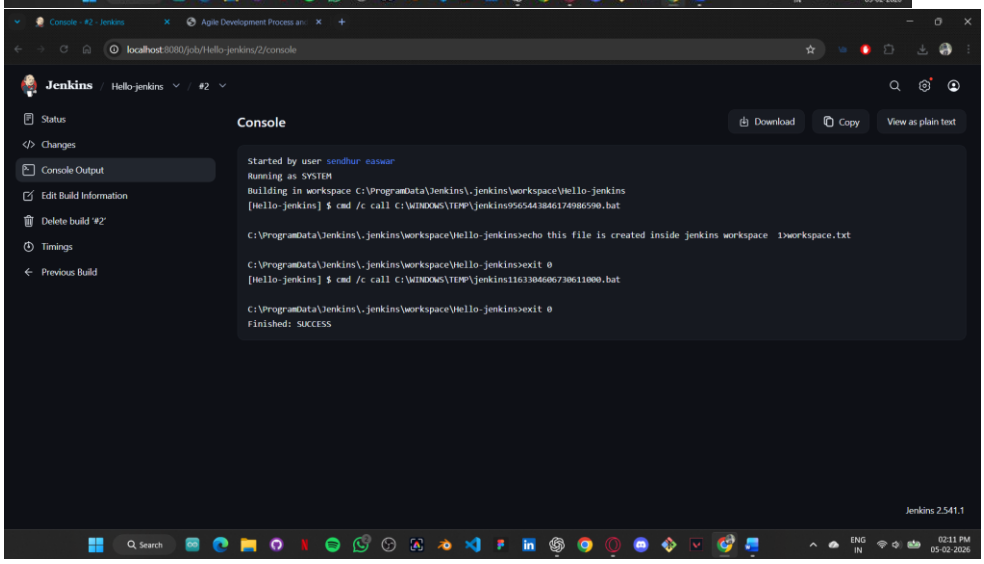
TASK 2: Create First Freestyle Job Objective: Create and run a Jenkins job Tasks: 1. Create a Freestyle project named Hello-Jenkins 2. Add a description 3. Add build step: o Execute shell / Windows batch command o Print "Hello Jenkins" 4. Build the job manually



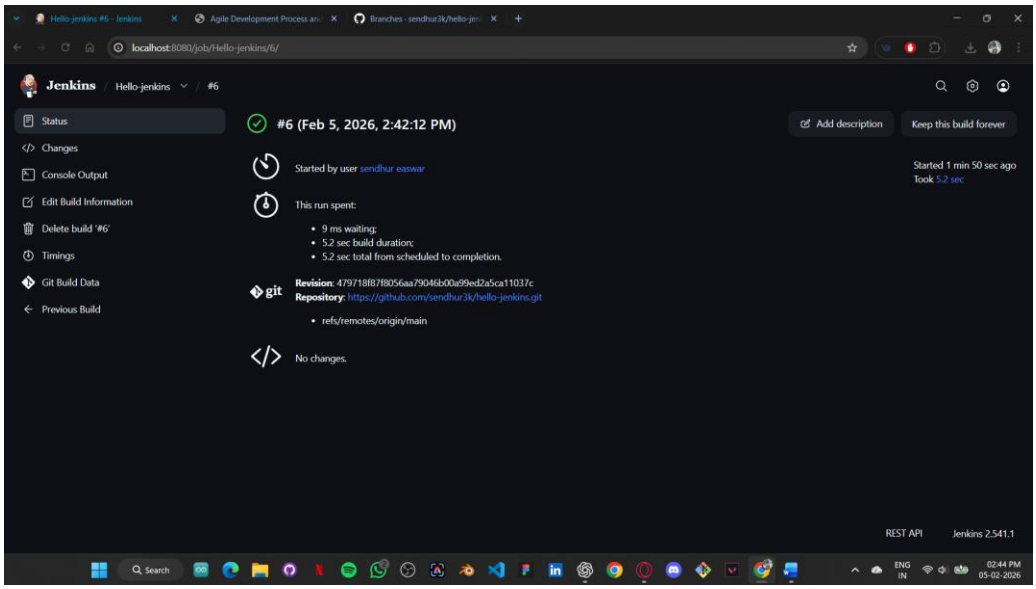


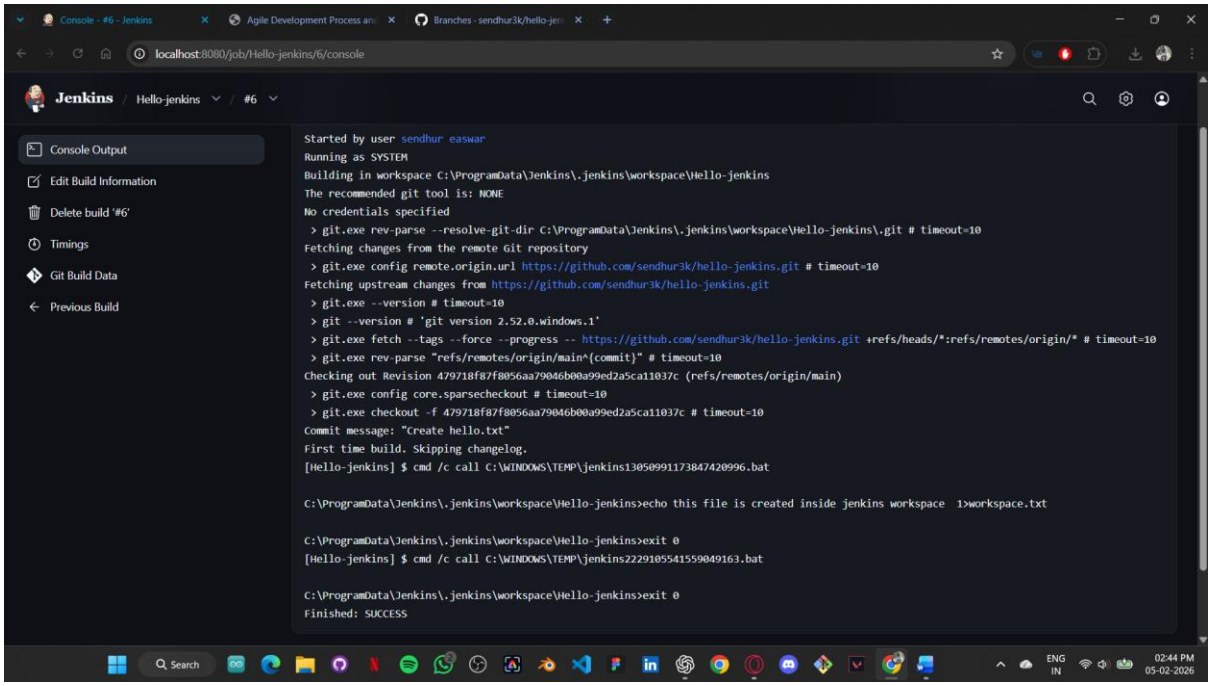
TASK 3: Jenkins Workspace & Commands Objective: Understand workspace usage Tasks: 1. Navigate to job workspace 2. Create a text file using build step 3. Display file contents in console Expected Output: File created inside workspace



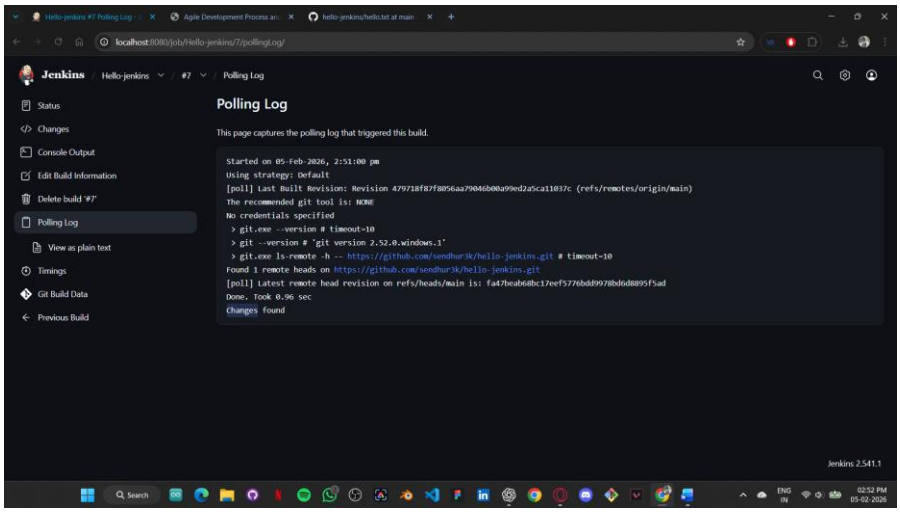


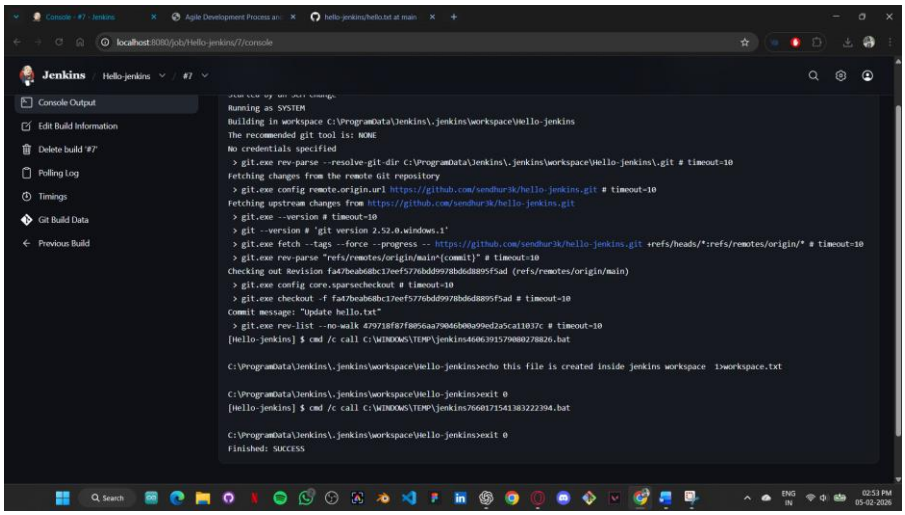
TASK 4: Git Integration Objective: Integrate Jenkins with GitHub Tasks: 1. Create a GitHub repository with sample code 2. Configure Git in Jenkins 3. Add Git repository URL in job 4. Build and verify code checkout Expected Output: • Source code visible in workspace



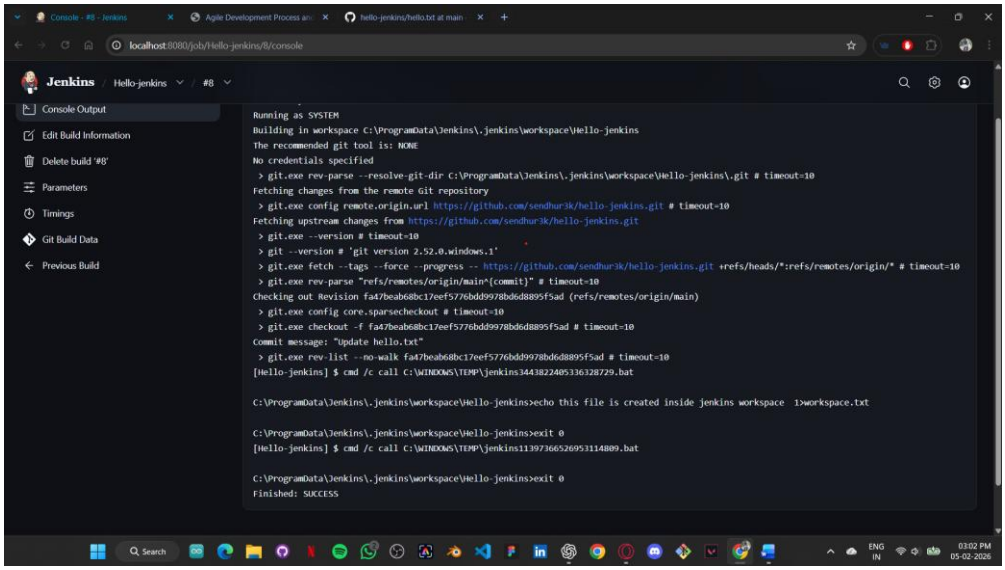


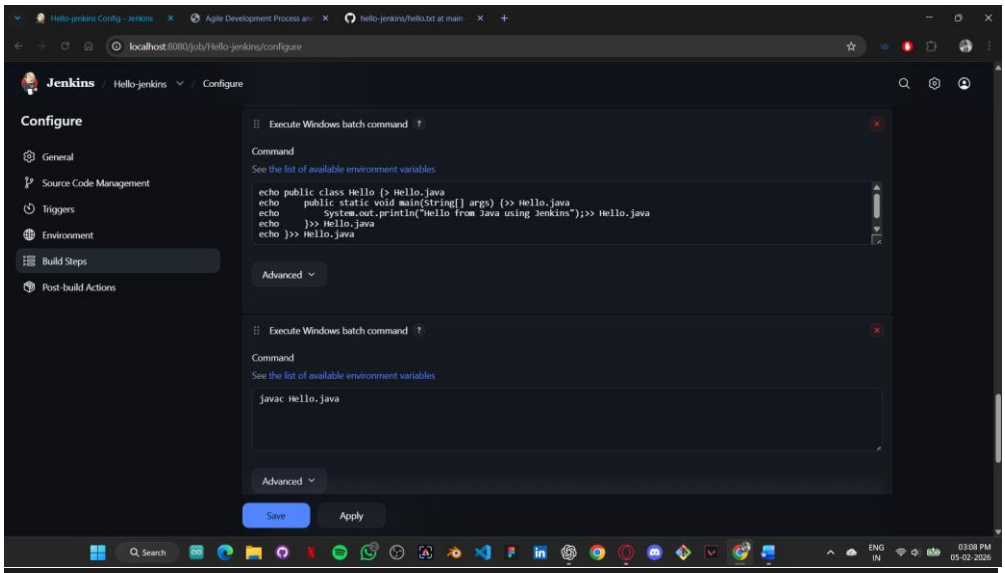
TASK 5: Poll SCM Trigger Objective: Automatically trigger builds on code change Tasks: 1. Enable Poll SCM 2. Set schedule: * * * * * 3. Modify GitHub file and commit 4. Observe automatic build Expected Output: • Build triggered without manual action

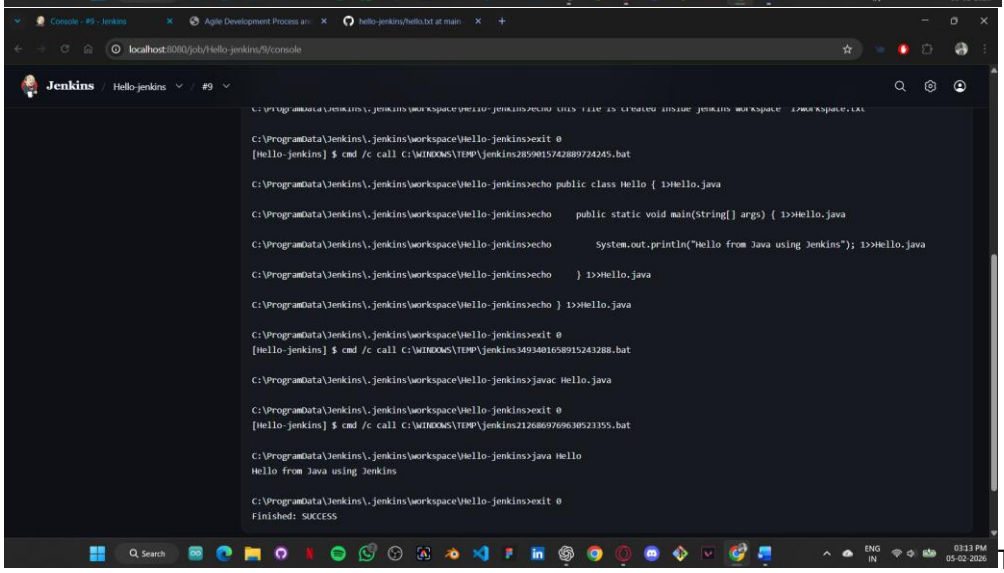




TASK 6: Parameterized Build Objective: Use parameters in Jenkins job Tasks: 1. Enable parameterized build 2. Add String parameter USERNAME 3. Print parameter value in build step Expected Output: • Console output showing parameter value

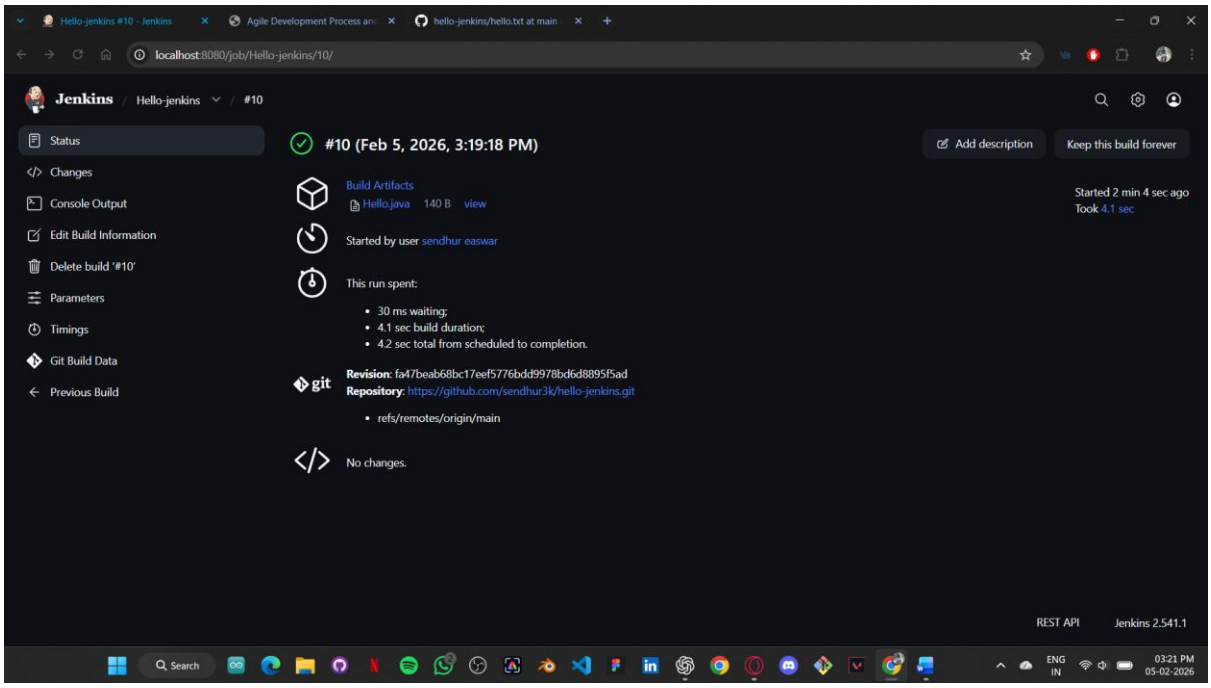




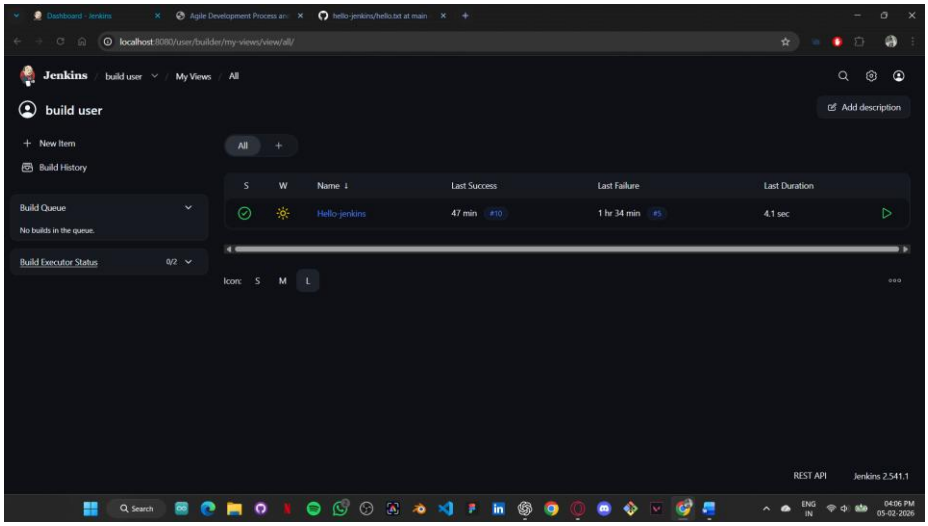


TASK 8: Archive

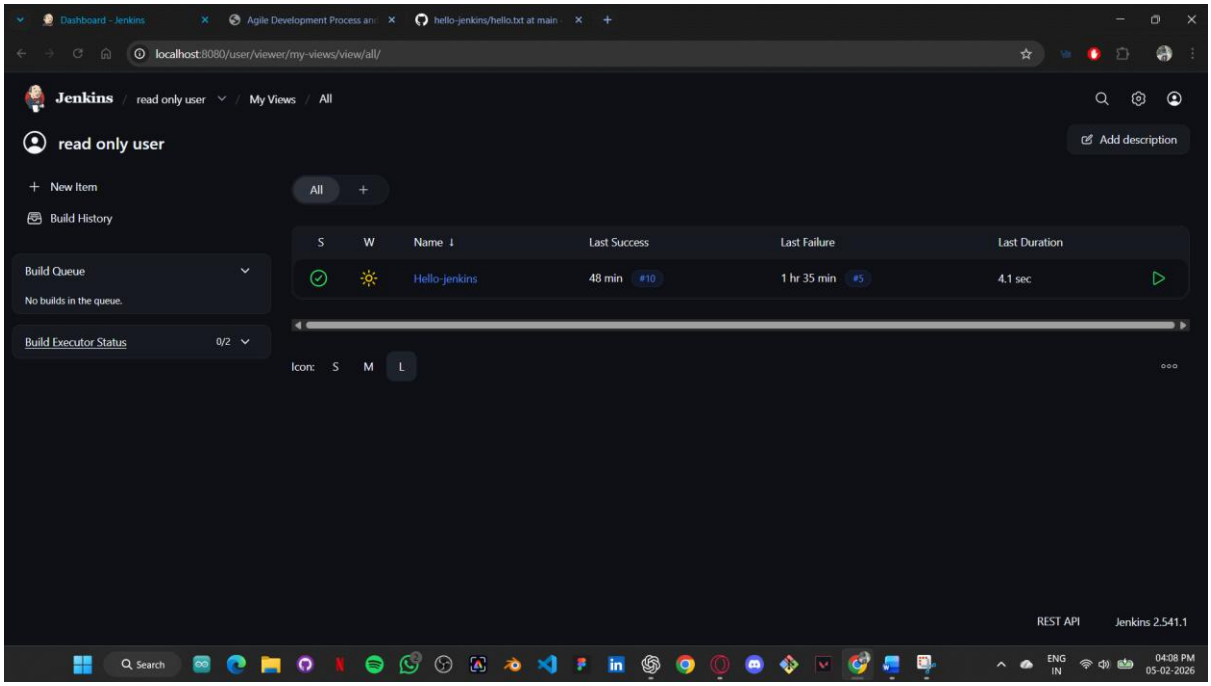
Artifacts Objective: Store build outputs Tasks: 1. Generate .class or .jar file 2. Archive artifacts in post-build action 3. Download artifact from Jenkins UI Expected Output: • Artifact available for download



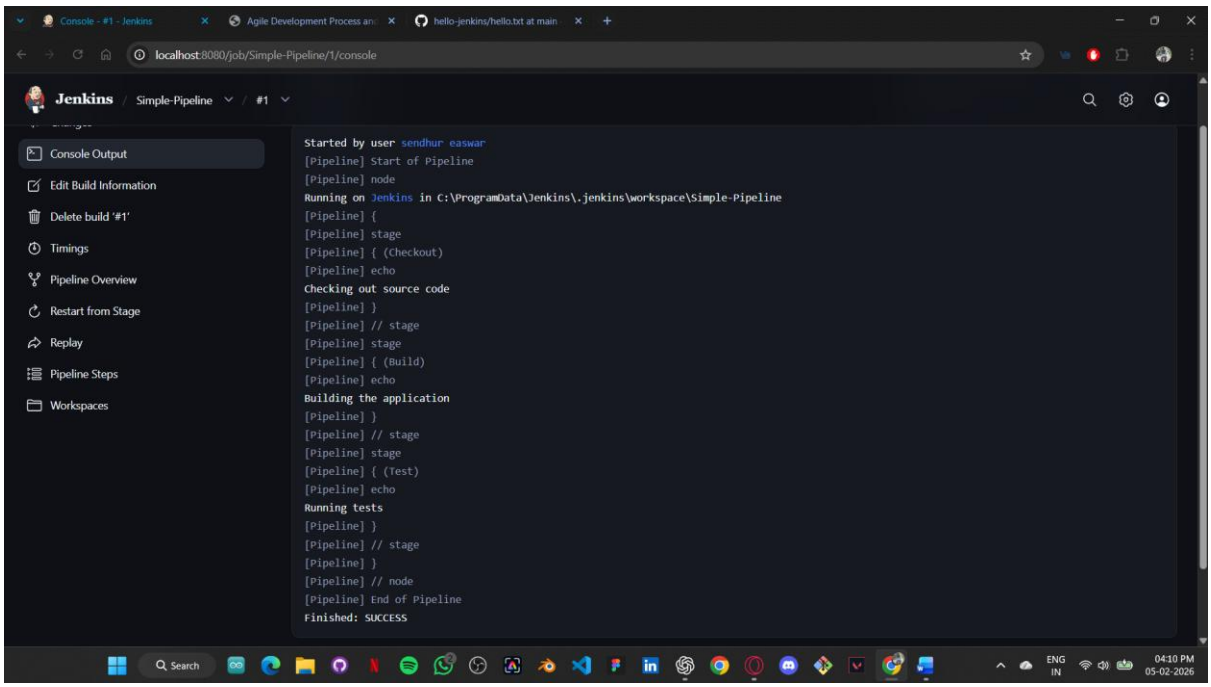
TASK 9: Users & Roles Objective: Manage Jenkins users Tasks: 1. Create two users 2. Assign read-only permission to one user 3. Assign build permission to another user Expected Output: • Permission differences verified



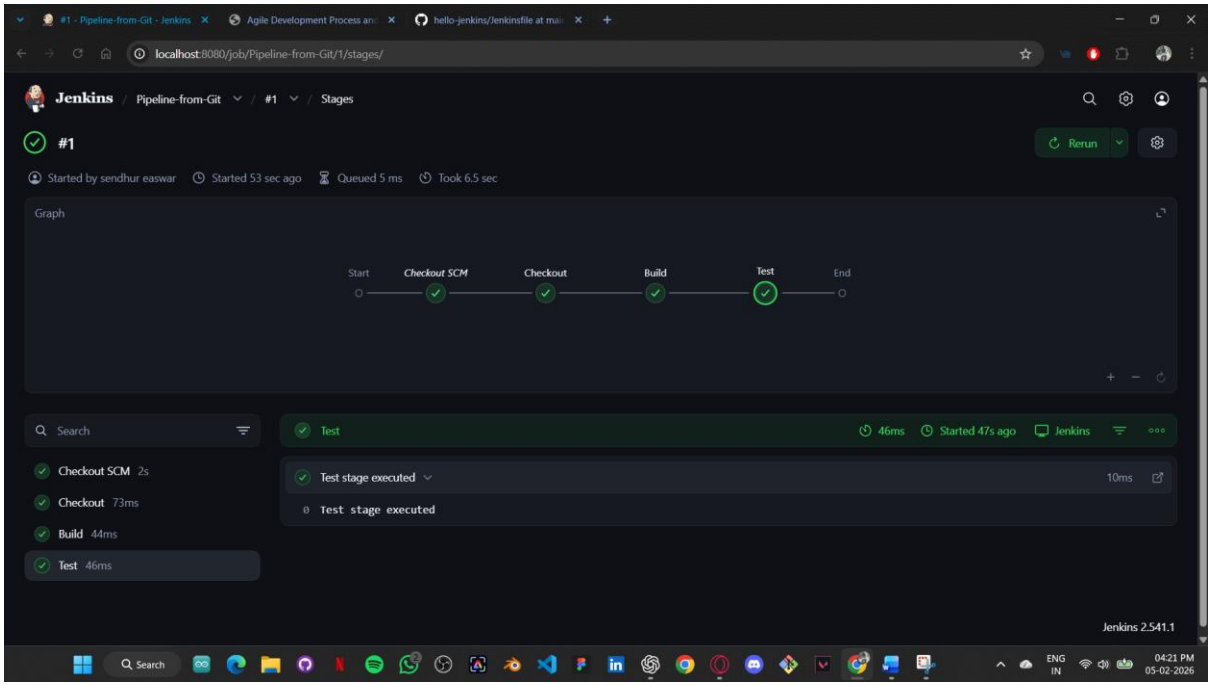
Viewer user



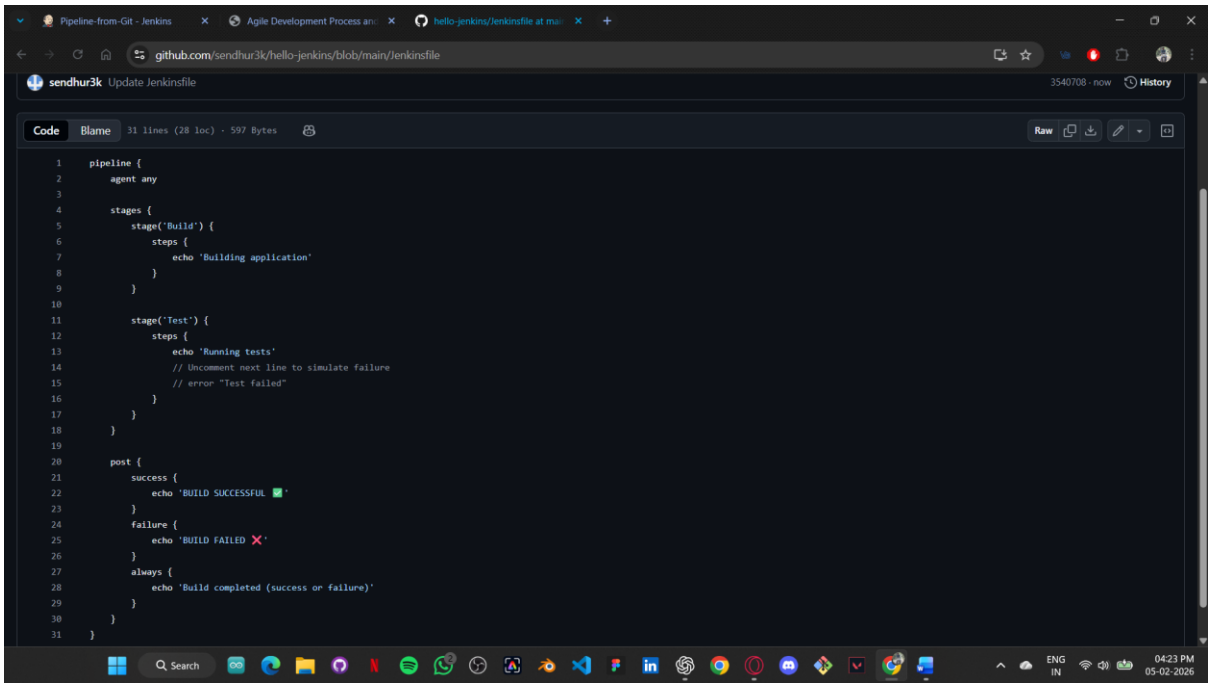
TASK 10: Simple Jenkins Pipeline Objective: Create basic pipeline Tasks: 1. Create Pipeline job 2. Write pipeline with stages: o Checkout o Build o Test 3. Run pipeline

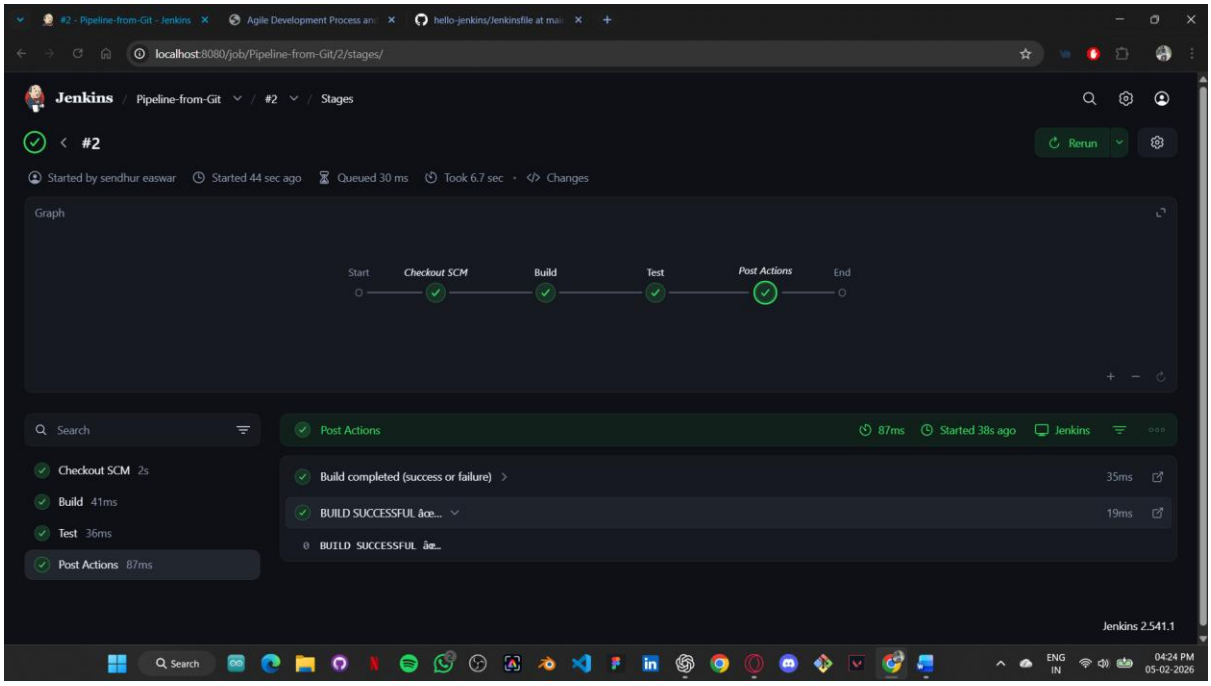


TASK 11: Jenkinsfile from Git Objective: Pipeline as Code Tasks: 1. Create Jenkinsfile in Git repo 2. Configure pipeline from SCM 3. Trigger build Expected Output: • Pipeline executed from Git

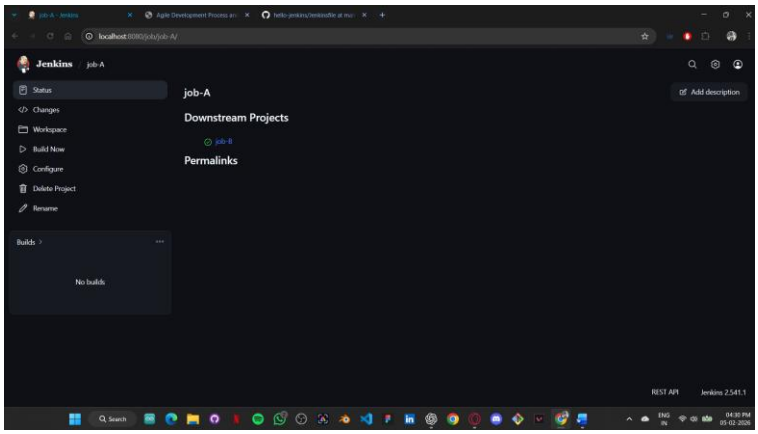


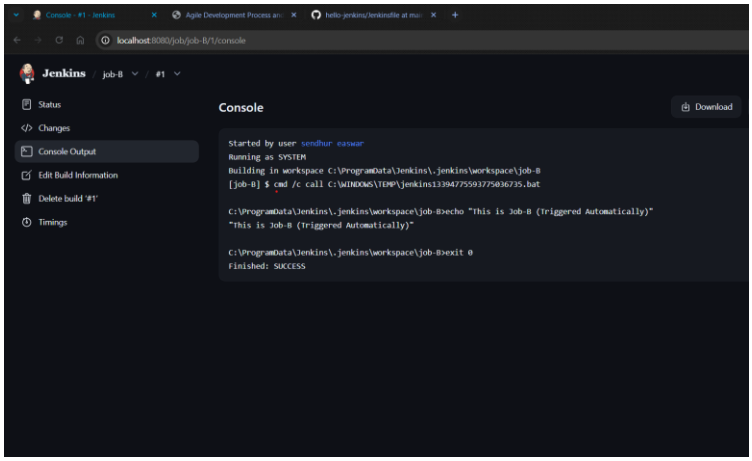
TASK 12: Post-Build Actions Objective: Handle build result Tasks: 1. Add post section 2. Print message on success/failure Expected Output: • Appropriate message displayed TASK 13: Trigger Job from Another Job Objective: Job chaining



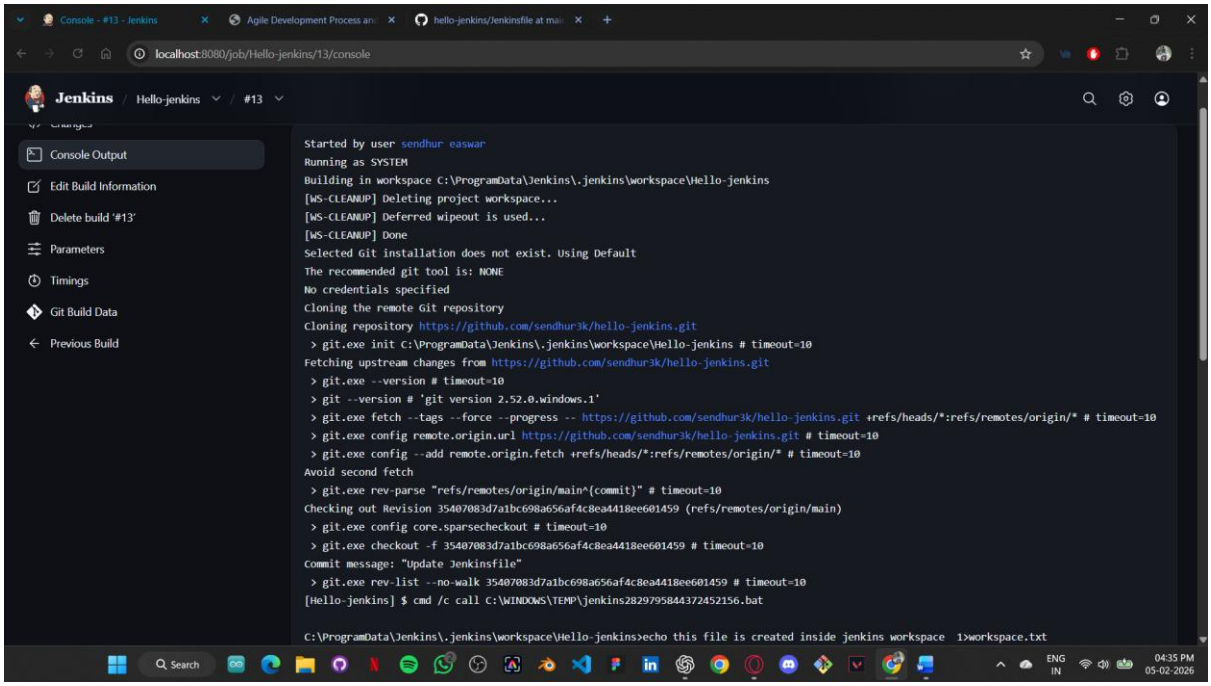


TASK 13: Trigger Job from Another Job Objective: Job chaining Tasks: 1. Create Job-A and Job-B 2. Configure Job-B to trigger after Job-A Expected Output: • Job-B triggered automatically

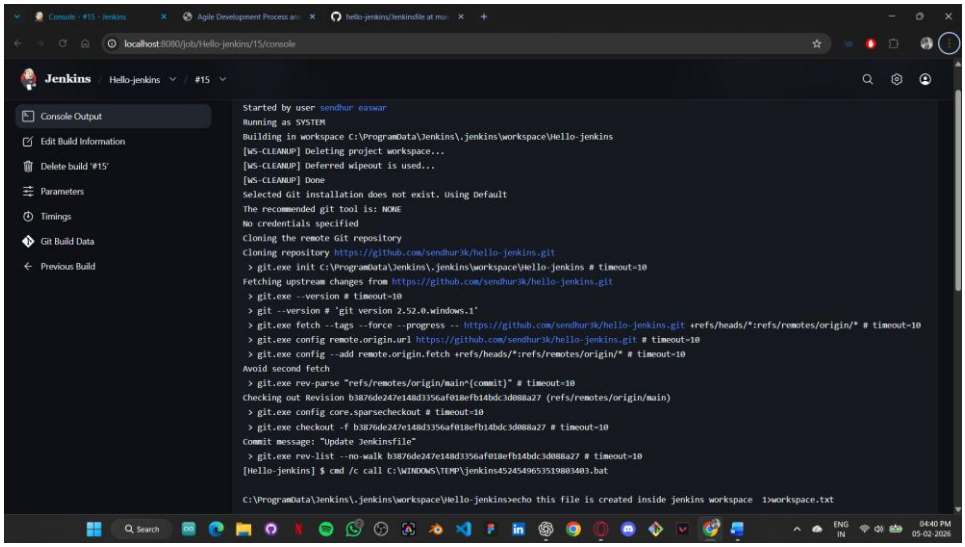




TASK 14: Workspace Cleanup Objective: Manage disk usage Tasks: 1. Install Workspace Cleanup plugin 2. Clean workspace before build Expected Output: • Workspace cleared before execution



TASK 15: Mini CI Project Objective: Implement basic CI flow Tasks: 1. Git commit → Jenkins build 2. Compile code 3. Archive artifacts 4. Fail build on error Expected Output: • Automated CI pipeline



Git Commit

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Jenkins Pipeline Triggered

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Compile Code

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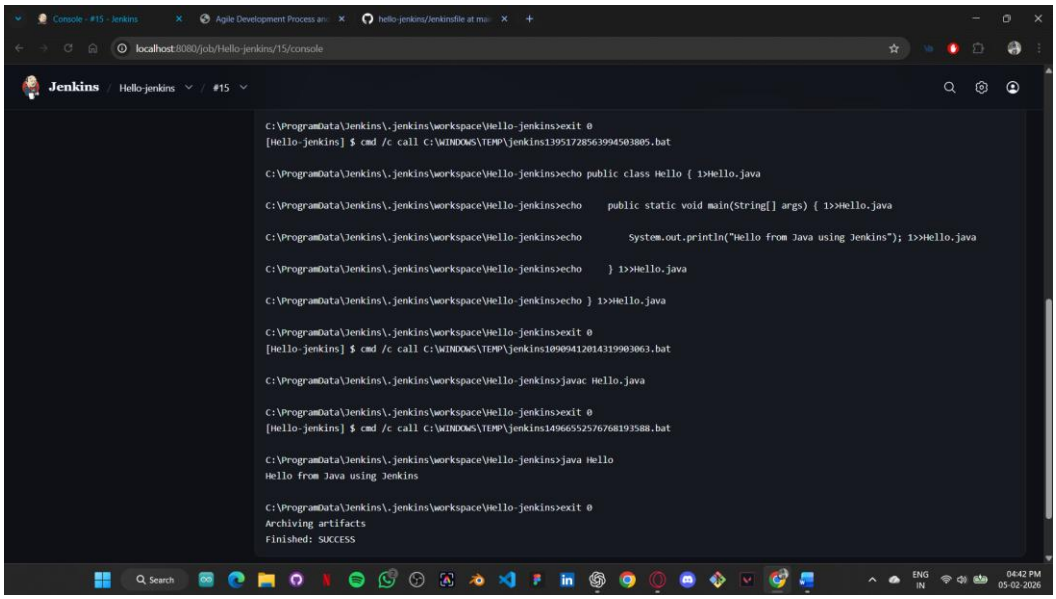
Archive Artifacts

↓

Fail Build if Error Occurs



CI USING TEKTON :

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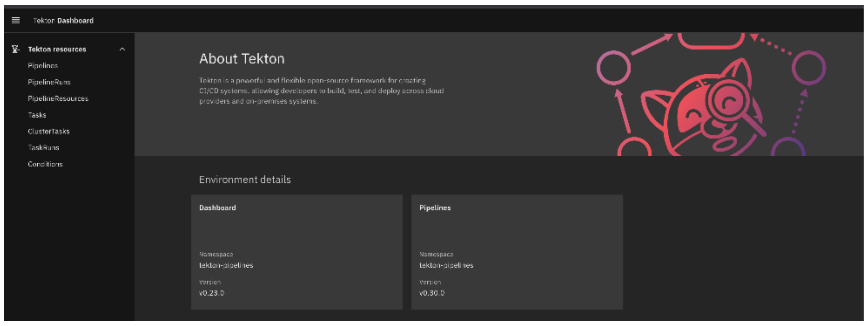

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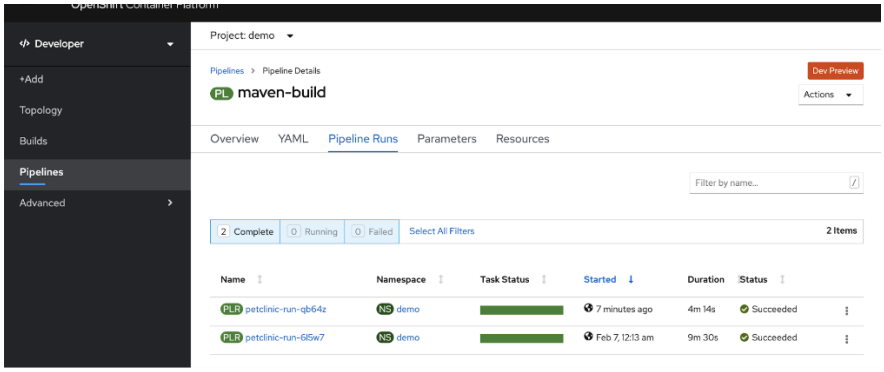

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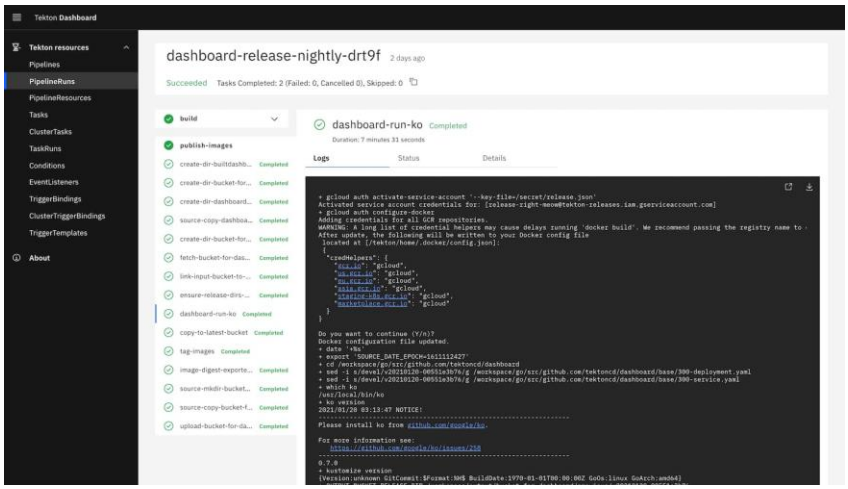

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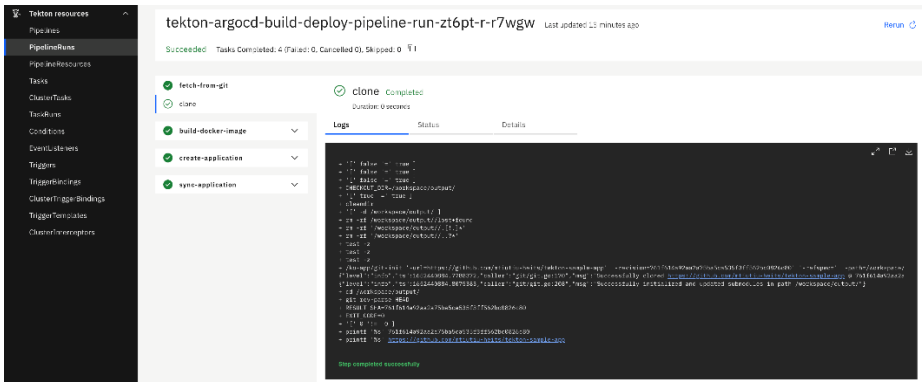

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